
LAZARD LEGACY MATH ACADEMY

3rd-4th Grade Math Complete

Addition • Subtraction • Multiplication • Division •
Fractions • Geometry

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Lazard Legacy Enterprises

"Dreams can come true. Better late than never."

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SECTION 1

Addition & Subtraction with Regrouping

Understanding Addition with Regrouping

When we add numbers, sometimes the digits in one column add up to more than 9. When this happens, we need to **regroup** — also called "carrying."

What does regrouping mean?

Regrouping means taking 10 ones and turning them into 1 ten, or taking 10 tens and turning them into 1 hundred. It's like trading 10 pennies for a dime!

Step-by-Step Process for Addition:

Step 1: Write the numbers in columns, lining up the ones, tens, and hundreds places.

Step 2: Start adding from the RIGHT side (ones place).

Step 3: If the sum in any column is 10 or more:

- Write the ONES digit of the sum below the line
- CARRY the TENS digit to the top of the next column

Step 4: Continue adding each column from right to left, including any carried numbers.

Important Tips:

- Always start from the ones (rightmost) column

- Write carried numbers small above the next column so you don't forget
- Double-check by estimating: round each number and see if your answer makes sense

Worked Examples: Addition with Regrouping

Example 1: $347 + 285$

Step 1: Line up the numbers: 347

+ 285

Step 2: Ones: $7 + 5 = 12$. Write 2, carry 1.

Step 3: Tens: $4 + 8 + 1(\text{carried}) = 13$. Write 3, carry 1.

Step 4: Hundreds: $3 + 2 + 1(\text{carried}) = 6$.

Answer: 632

Example 2: $568 + 274$

Ones: $8 + 4 = 12$. Write 2, carry 1.

Tens: $6 + 7 + 1 = 14$. Write 4, carry 1.

Hundreds: $5 + 2 + 1 = 8$.

Answer: 842

Example 3: $493 + 318$

Ones: $3 + 8 = 11$. Write 1, carry 1.

Tens: $9 + 1 + 1 = 11$. Write 1, carry 1.

Hundreds: $4 + 3 + 1 = 8$.

Answer: 811

Example 4: $756 + 189$

Ones: $6 + 9 = 15$. Write 5, carry 1.

Tens: $5 + 8 + 1 = 14$. Write 4, carry 1.

Hundreds: $7 + 1 + 1 = 9$.

Answer: 945

Example 5: $624 + 397$

Ones: $4 + 7 = 11$. Write 1, carry 1.

Tens: $2 + 9 + 1 = 12$. Write 2, carry 1.

Hundreds: $6 + 3 + 1 = 10$. Write 0, carry 1.

Thousands: 1.

Answer: 1,021

More Addition Examples

Example 6: $899 + 456$

Ones: $9 + 6 = 15$. Write 5, carry 1.

Tens: $9 + 5 + 1 = 15$. Write 5, carry 1.

Hundreds: $8 + 4 + 1 = 13$. Write 3, carry 1.

Thousands: 1.

Answer: 1,355

Example 7: $267 + 548$

Ones: $7 + 8 = 15$. Write 5, carry 1.

Tens: $6 + 4 + 1 = 11$. Write 1, carry 1.

Hundreds: $2 + 5 + 1 = 8$.

Answer: 815

Example 8: $439 + 372$

Ones: $9 + 2 = 11$. Write 1, carry 1.

Tens: $3 + 7 + 1 = 11$. Write 1, carry 1.

Hundreds: $4 + 3 + 1 = 8$.

Answer: 811

Example 9: $185 + 967$

Ones: $5 + 7 = 12$. Write 2, carry 1.

Tens: $8 + 6 + 1 = 15$. Write 5, carry 1.

Hundreds: $1 + 9 + 1 = 11$. Write 1, carry 1.

Thousands: 1.

Answer: 1,152

Example 10: $773 + 658$

Ones: $3 + 8 = 11$. Write 1, carry 1.

Tens: $7 + 5 + 1 = 13$. Write 3, carry 1.

Hundreds: $7 + 6 + 1 = 14$. Write 4, carry 1.

Thousands: 1.

Answer: 1,431

Understanding Subtraction with Regrouping

Sometimes when we subtract, the top digit is **smaller** than the bottom digit. We can't take away more than we have! So we need to **borrow** (regroup) from the next column.

What does borrowing mean?

Borrowing means taking 1 ten and breaking it into 10 ones, or taking 1 hundred and breaking it into 10 tens. It's like breaking a dollar into 10 dimes!

Step-by-Step Process for Subtraction:

Step 1: Write the larger number on TOP, smaller number on bottom. Line up place values.

Step 2: Start from the RIGHT (ones place). Can you subtract? If YES, subtract normally.

Step 3: If the top digit is SMALLER than the bottom digit:

- Look at the column to the LEFT
- Reduce that digit by 1 (you're borrowing from it)
- Add 10 to the current column's top digit
- Now subtract!

Step 4: Continue moving left through each column.

Special Case — Borrowing Across Zeros:

If the next column is 0, you need to keep going left until you find a non-zero digit to borrow from. Example: $500 - 237$ requires borrowing across the zero in the tens place.

Worked Examples: Subtraction with Regrouping

Example 1: $532 - 287$

Ones: $2 < 7$, so borrow from tens. Tens goes from $3 \rightarrow 2$, ones becomes 12.

$$12 - 7 = 5.$$

Tens: $2 < 8$, so borrow from hundreds. Hundreds goes from $5 \rightarrow 4$, tens becomes 12.

$$12 - 8 = 4.$$

$$\text{Hundreds: } 4 - 2 = 2.$$

Answer: 245

Example 2: $804 - 356$

Ones: $4 < 6$, borrow from tens. But tens is 0!

Borrow from hundreds: $8 \rightarrow 7$, tens becomes 10, then tens $10 \rightarrow 9$, ones becomes 14.

$$14 - 6 = 8.$$

$$\text{Tens: } 9 - 5 = 4.$$

$$\text{Hundreds: } 7 - 3 = 4.$$

Answer: 448

Example 3: $700 - 463$

Ones: $0 < 3$, need to borrow. Tens is 0, so borrow from hundreds.

Hundreds: $7 \rightarrow 6$, tens becomes 10. Then tens: $10 \rightarrow 9$, ones becomes 10.

$$10 - 3 = 7.$$

$$\text{Tens: } 9 - 6 = 3.$$

$$\text{Hundreds: } 6 - 4 = 2.$$

Answer: 237

Example 4: 921 - 548

Ones: $1 < 8$, borrow from tens: tens $2 \rightarrow 1$, ones becomes 11.

$$11 - 8 = 3.$$

Tens: $1 < 4$, borrow from hundreds: hundreds $9 \rightarrow 8$, tens becomes 11.

$$11 - 4 = 7.$$

$$\text{Hundreds: } 8 - 5 = 3.$$

Answer: 373

Example 5: 615 - 278

Ones: $5 < 8$, borrow from tens: tens $1 \rightarrow 0$, ones becomes 15.

$$15 - 8 = 7.$$

Tens: $0 < 7$, borrow from hundreds: hundreds $6 \rightarrow 5$, tens becomes 10.

$$10 - 7 = 3.$$

$$\text{Hundreds: } 5 - 2 = 3.$$

Answer: 337

More Subtraction Examples

Example 6: $462 - 185$

Ones: $2 < 5$, borrow: tens $6 \rightarrow 5$, ones $12 - 5 = 7$.

Tens: $5 < 8$, borrow: hundreds $4 \rightarrow 3$, tens $15 - 8 = 7$.

Hundreds: $3 - 1 = 2$.

Answer: 277

Example 7: $1000 - 637$

Ones: $0 < 7$, borrow across zeros.

Thousands $1 \rightarrow 0$, hundreds $0 \rightarrow 9$, tens $0 \rightarrow 9$, ones $0 \rightarrow 10$.

$10 - 7 = 3$, $9 - 3 = 6$, $9 - 6 = 3$.

Answer: 363

Example 8: $853 - 496$

Ones: $3 < 6$, borrow: tens $5 \rightarrow 4$, ones $13 - 6 = 7$.

Tens: $4 < 9$, borrow: hundreds $8 \rightarrow 7$, tens $14 - 9 = 5$.

Hundreds: $7 - 4 = 3$.

Answer: 357

Example 9: $741 - 365$

Ones: $1 < 5$, borrow: tens $4 \rightarrow 3$, ones $11 - 5 = 6$.

Tens: $3 < 6$, borrow: hundreds $7 \rightarrow 6$, tens $13 - 6 = 7$.

Hundreds: $6 - 3 = 3$.

Answer: 376

Example 10: $500 - 234$

Ones: $0 < 4$, borrow. Tens is 0, borrow from hundreds.

Hundreds $5 \rightarrow 4$, tens $0 \rightarrow 9$ (after giving 1 to ones), ones $10 - 4 = 6$.

Actually: $5 \rightarrow 4$, tens $10 \rightarrow 9$, ones 10. $10 - 4 = 6$, $9 - 3 = 6$, $4 - 2 = 2$.

Answer: 266

Practice Problems: Addition & Subtraction

Directions: Solve each problem. Show your work on the lines provided.

Addition (Problems 1-10):

1. $804 + 264 =$ _____

2. $175 + 431 =$ _____

3. $400 + 378 =$ _____

4. $292 + 254 =$ _____

5. $842 + 708 =$ _____

6. $239 + 754 =$ _____

7. $582 + 182 =$ _____

8. $180 + 245 =$ _____

9. $373 + 388 =$ _____

10. $667 + 766 =$ _____

Subtraction (Problems 11-20):

11. $427 - 293 =$ _____

12. $603 - 516 =$ _____

13. $958 - 579 =$ _____

14. $625 - 379 =$ _____

15. $684 - 564 =$ _____

16. $406 - 344 =$ _____

17. $563 - 507 =$ _____

18. $832 - 498 =$ _____

19. $684 - 229 =$ _____

20. $620 - 540 =$ _____

SECTION 2

Multiplication

Understanding Multiplication

Multiplication is a fast way to add the same number many times.

When we write 3×4 , we mean "3 groups of 4" which is the same as $4 + 4 + 4 = 12$.

Key Vocabulary:

- **Factors** — the numbers being multiplied (in 3×4 , both 3 and 4 are factors)
- **Product** — the answer to a multiplication problem (12 is the product of 3×4)
- **Array** — objects arranged in equal rows and columns to show multiplication

Multiplication as Repeated Addition:

- $5 \times 3 = 3 + 3 + 3 + 3 + 3 = 15$ (five groups of three)
- $4 \times 6 = 6 + 6 + 6 + 6 = 24$ (four groups of six)
- $7 \times 2 = 2 + 2 + 2 + 2 + 2 + 2 + 2 = 14$ (seven groups of two)

The Commutative Property (Order Property):

The order of factors doesn't change the product!

- $3 \times 4 = 4 \times 3 = 12$
- $6 \times 8 = 8 \times 6 = 48$

This means if you know $3 \times 7 = 21$, you also know $7 \times 3 = 21$!

The Zero Property:

Any number multiplied by zero equals zero: $5 \times 0 = 0$

The Identity Property:

Any number multiplied by one equals itself: $8 \times 1 = 8$

Multiplication Table Tips & Tricks

×2 Table: Just double the number! $2 \times 7 = 7 + 7 = 14$

×3 Table: The digits of multiples of 3 always add up to 3, 6, or 9.

3, 6, 9, $12(1+2=3)$, $15(1+5=6)$, $18(1+8=9)$, $21(2+1=3)$...

×4 Table: Double, then double again! 4×6 : first $2 \times 6 = 12$, then $2 \times 12 = 24$

×5 Table: Count by 5s! Products always end in 0 or 5.

5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60

×9 Table — The Finger Trick:

Hold up 10 fingers. Put down the finger matching the number you multiply by.

Fingers to the LEFT = tens digit. Fingers to the RIGHT = ones digit.

Example: 9×7 : put down finger 7. Left: 6 fingers. Right: 3 fingers. Answer: 63!

×9 Table — The Pattern:

$9 \times 1 = 09$, $9 \times 2 = 18$, $9 \times 3 = 27$, $9 \times 4 = 36$, $9 \times 5 = 45$...

Notice: tens digit goes UP by 1, ones digit goes DOWN by 1. Digits always add to 9!

×10 Table: Just add a zero! $10 \times 7 = 70$

×11 Table (1-9): Repeat the digit! $11 \times 3 = 33$, $11 \times 7 = 77$

×12 Table: Think of it as ×10 plus ×2. Example: $12 \times 7 = 70 + 14 = 84$

Worked Examples: Multiplication

Example 1: $6 \times 7 = ?$

$6 \times 7 = 42$. Memory trick: "Six times seven = forty-two"
or think 6,7,8 $\rightarrow 6 \times 7 = 42$

Example 2: $8 \times 9 = ?$

$8 \times 9 = 72$. Finger trick: hold down finger 8, see 7 and 2. Also: $8 \times 9 = 8 \times 10 - 8 = 80 - 8 = 72$

Example 3: $12 \times 4 = ?$

$12 \times 4 = 48$. Break it up: $10 \times 4 = 40$, $2 \times 4 = 8$, $40 + 8 = 48$

Example 4: $7 \times 8 = ?$

$7 \times 8 = 56$. Memory trick: "5, 6, 7, 8" $\rightarrow 56 = 7 \times 8$

Example 5: $9 \times 6 = ?$

$9 \times 6 = 54$. Nine trick: tens = $6 - 1 = 5$, ones = $9 - 5 = 4$,
answer = 54

Example 6: $11 \times 7 = ?$

$11 \times 7 = 77$. Pattern: for single digits, $11 \times$ just repeats the digit!

Example 7: $3 \times 12 = ?$

$3 \times 12 = 36$. Think: $3 \times 10 = 30$, $3 \times 2 = 6$, $30 + 6 = 36$

Example 8: $8 \times 8 = ?$

$8 \times 8 = 64$. Perfect square! Say: "I ate and I ate till I was sick on the floor" ($8 \times 8 = 64$)

Example 9: $5 \times 9 = ?$

$5 \times 9 = 45$. Count by 5s nine times:
5, 10, 15, 20, 25, 30, 35, 40, 45

Example 10: $7 \times 12 = ?$

$7 \times 12 = 84$. Break it: $7 \times 10 = 70$, $7 \times 2 = 14$, $70 + 14 = 84$

Practice Problems: Multiplication

Directions: Solve each multiplication problem.

1. $3 \times 7 =$ _____

2. $4 \times 8 =$ _____

3. $6 \times 9 =$ _____

4. $7 \times 7 =$ _____

5. $8 \times 6 =$ _____

6. $9 \times 4 =$ _____

7. $5 \times 8 =$ _____

8. $12 \times 3 =$ _____

9. $11 \times 6 =$ _____

10. $4 \times 9 =$ _____

Multiplication Practice (continued)

11. $6 \times 6 =$ _____

12. $7 \times 9 =$ _____

13. $8 \times 12 =$ _____

14. $9 \times 9 =$ _____

15. $5 \times 7 =$ _____

16. $3 \times 11 =$ _____

17. $6 \times 8 =$ _____

18. $7 \times 4 =$ _____

19. $12 \times 5 =$ _____

20. $8 \times 7 =$ _____

SECTION 3

Division

Understanding Division

Division means splitting a number into **equal groups**. It's the opposite (inverse) of multiplication!

Key Vocabulary:

- **Dividend** — the number being divided (the total you're splitting up)
- **Divisor** — the number you divide by (how many groups, or size of each group)
- **Quotient** — the answer to a division problem
- **Remainder** — what's left over when division isn't exact

In $42 \div 6 = 7$: 42 is the dividend, 6 is the divisor, 7 is the quotient.

Division and Multiplication Connection:

These fact families are connected:

- If $6 \times 7 = 42$, then $42 \div 7 = 6$ AND $42 \div 6 = 7$
- If $8 \times 5 = 40$, then $40 \div 5 = 8$ AND $40 \div 8 = 5$

To solve a division problem, ask: "What number times the divisor gives me the dividend?"

Division with Remainders:

Sometimes numbers don't divide evenly. The leftover amount is the **remainder**.

Example: $17 \div 5 = 3 \text{ R}2$ (because $5 \times 3 = 15$, and $17 - 15 = 2$ left over)

Check your work: (Quotient $\times$ Divisor) + Remainder = Dividend

Check: $(3 \times 5) + 2 = 15 + 2 = 17 \checkmark$

Division Rules to Remember:

- Any number $\div 1 =$ itself ($20 \div 1 = 20$)
- Any number $\div$ itself $= 1$ ($7 \div 7 = 1$)
- Zero $\div$ any number $= 0$ ($0 \div 5 = 0$)
- You CANNOT divide by zero! ($5 \div 0 =$ undefined)

Worked Examples: Division

Example 1: $36 \div 4 = ?$

Think: $4 \times ? = 36$. Since $4 \times 9 = 36$, the answer is 9.

Example 2: $56 \div 8 = ?$

Think: $8 \times ? = 56$. Since $8 \times 7 = 56$, the answer is 7.

Example 3: $45 \div 9 = ?$

Think: $9 \times ? = 45$. Since $9 \times 5 = 45$, the answer is 5.

Example 4: $72 \div 6 = ?$

Think: $6 \times ? = 72$. Since $6 \times 12 = 72$, the answer is 12.

Example 5: $84 \div 7 = ?$

Think: $7 \times ? = 84$. Since $7 \times 12 = 84$, the answer is 12.

Example 6: $25 \div 4 = ?$

$4 \times 6 = 24$ (closest without going over). $25 - 24 = 1$.

Answer: 6 R1.

Example 7: $47 \div 5 = ?$

$5 \times 9 = 45$ (closest without going over). $47 - 45 = 2$.

Answer: 9 R2.

Example 8: $67 \div 8 = ?$

$8 \times 8 = 64$ (closest without going over). $67 - 64 = 3$.

Answer: 8 R3.

Example 9: $100 \div 12 = ?$

$12 \times 8 = 96$ (closest without going over). $100 - 96 = 4$.

Answer: 8 R4.

Example 10: $55 \div 6 = ?$

$6 \times 9 = 54$ (closest without going over). $55 - 54 = 1$.

Answer: 9 R1.

Practice Problems: Division

Directions: Solve each division problem. Write the remainder if there is one.

1. $48 \div 6 =$ _____

2. $63 \div 7 =$ _____

3. $81 \div 9 =$ _____

4. $54 \div 6 =$ _____

5. $96 \div 8 =$ _____

6. $72 \div 9 =$ _____

7. $108 \div 12 =$ _____

8. $44 \div 4 =$ _____

9. $60 \div 5 =$ _____

10. $77 \div 7 =$ _____

Division Practice (continued)

11. $29 \div 3 =$ _____

12. $37 \div 4 =$ _____

13. $50 \div 6 =$ _____

14. $65 \div 8 =$ _____

15. $43 \div 5 =$ _____

16. $58 \div 7 =$ _____

17. $83 \div 9 =$ _____

18. $95 \div 11 =$ _____

19. $47 \div 6 =$ _____

20. $71 \div 8 =$ _____

SECTION 4

Fractions Introduction

What Is a Fraction?

A **fraction** represents a PART of a WHOLE. When we cut something into equal pieces, each piece is a fraction of the whole.

Imagine cutting a pizza into 8 equal slices. If you eat 3 slices, you ate $\frac{3}{8}$ of the pizza!

Parts of a Fraction:

3 ← NUMERATOR (top): how many parts you HAVE

8 ← DENOMINATOR (bottom): total equal parts in the whole

Reading Fractions:

- $\frac{1}{2}$ = "one half"
- $\frac{1}{3}$ = "one third"
- $\frac{3}{4}$ = "three fourths" or "three quarters"
- $\frac{5}{8}$ = "five eighths"

Types of Fractions:

- **Proper fraction:** numerator < denominator (less than 1 whole). Ex: $\frac{3}{4}$, $\frac{1}{2}$, $\frac{5}{8}$
- **Improper fraction:** numerator $\geq$ denominator (equal to or greater than 1 whole). Ex: $\frac{5}{3}$, $\frac{9}{4}$

- **Mixed number:** a whole number plus a fraction. Ex: $1 \frac{1}{2}$, $2 \frac{3}{4}$

Equivalent Fractions & Comparing

Equivalent Fractions:

Fractions that look different but represent the SAME amount.

- $1/2 = 2/4 = 3/6 = 4/8$ (all are exactly half!)

How to make equivalent fractions:

Multiply (or divide) BOTH the numerator AND denominator by the same number.

- $1/3 \times 2/2 = 2/6$ (multiplied both by 2)
- $2/3 \times 4/4 = 8/12$ (multiplied both by 4)

Simplifying Fractions:

Divide both parts by their Greatest Common Factor (GCF).

- $4/8$: GCF of 4 and 8 is 4. Divide both by 4: $1/2$
- $6/9$: GCF of 6 and 9 is 3. Divide both by 3: $2/3$

Comparing Fractions:

Rule 1 — Same denominator: Just compare numerators!

- $3/8$ vs $5/8 \rightarrow 3 < 5$, so $3/8 < 5/8$

Rule 2 — Same numerator: LARGER denominator = SMALLER fraction!

- $1/4$ vs $1/6 \rightarrow 4 < 6$, but $1/4 > 1/6$ (cutting into fewer pieces means bigger pieces)

Rule 3 — Different: Find a common denominator, then compare.

- Compare $\frac{2}{3}$ and $\frac{3}{5}$: LCD=15. $\frac{2}{3}=\frac{10}{15}$, $\frac{3}{5}=\frac{9}{15}$. So $\frac{2}{3} > \frac{3}{5}$

Worked Examples: Fractions

Example 1: Write a fraction: 2 out of 5 equal parts

Answer: $\frac{2}{5}$ – numerator is 2 (parts we have),
denominator is 5 (total parts)

Example 2: 3 of 8 squares are shaded. What fraction is shaded?

Answer: $\frac{3}{8}$ of the squares are shaded

Example 3: Find an equivalent fraction for $\frac{1}{2}$ with denominator 6

Answer: Multiply both by 3: $\frac{1 \times 3}{2 \times 3} = \frac{3}{6}$

Example 4: Find an equivalent fraction for $\frac{2}{3}$ with denominator 12

Answer: Multiply both by 4: $\frac{2 \times 4}{3 \times 4} = \frac{8}{12}$

Example 5: Compare: $\frac{2}{5}$ and $\frac{3}{5}$

Answer: Same denominator $\rightarrow$ compare numerators: $2 < 3$,
so $\frac{2}{5} < \frac{3}{5}$

Example 6: Compare: $\frac{1}{4}$ and $\frac{1}{3}$

Answer: Same numerator $\rightarrow$ larger denominator is smaller:
 $\frac{1}{4} < \frac{1}{3}$

Example 7: Convert $\frac{7}{4}$ to a mixed number

Answer: $7 \div 4 = 1$ remainder 3, so $\frac{7}{4} = 1 \frac{3}{4}$

Example 8: Convert $2 \frac{1}{3}$ to an improper fraction

Answer: $(2 \times 3) + 1 = 7$, so $2 \frac{1}{3} = \frac{7}{3}$

Example 9: Simplify $\frac{4}{8}$

Answer: GCF of 4 and 8 is 4. Divide both: $4 \div 4 / 8 \div 4 = \frac{1}{2}$

Example 10: What fraction of a dollar is 25 cents?

Answer: $\frac{25}{100}$ – simplify by dividing both by 25 $\rightarrow \frac{1}{4}$

Practice Problems: Fractions

Directions: Solve each fraction problem.

1. Write a fraction for: 5 out of 9 equal parts
2. Write a fraction for: 7 out of 10 equal parts
3. What is the numerator in $\frac{3}{7}$?
4. What is the denominator in $\frac{5}{12}$?
5. Find an equivalent fraction for $\frac{1}{3}$ with denominator 9
6. Find an equivalent fraction for $\frac{2}{5}$ with denominator 15
7. Find an equivalent fraction for $\frac{3}{4}$ with denominator 20
8. Simplify: $\frac{6}{8}$
9. Simplify: $\frac{4}{12}$
10. Simplify: $\frac{10}{15}$
11. Compare using $<$ or $>$: $\frac{3}{7}$ ____ $\frac{5}{7}$

12. Compare using $<$ or $>$: $\frac{1}{6}$ ____ $\frac{1}{4}$
13. Compare using $<$ or $>$: $\frac{2}{3}$ ____ $\frac{3}{5}$ (hint: find common denominator)
14. Convert to a mixed number: $\frac{9}{4}$
15. Convert to a mixed number: $\frac{11}{3}$
16. Convert to an improper fraction: $3 \frac{1}{2}$
17. Convert to an improper fraction: $2 \frac{2}{5}$
18. What fraction of 1 hour is 15 minutes?
19. What fraction of 1 foot (12 inches) is 4 inches?
20. You eat 3 slices of pizza cut into 8 slices. What fraction is LEFT?

SECTION 5

Basic Geometry

Shapes You Should Know

2D (Flat) Shapes:

- **Triangle** — 3 sides, 3 angles. Types: equilateral (all sides equal), isosceles (2 equal), scalene (no equal sides).
- **Square** — 4 EQUAL sides, 4 right angles (90°). A special rectangle!
- **Rectangle** — 4 sides, opposite sides are equal, 4 right angles (90°).
- **Parallelogram** — 4 sides, opposite sides are parallel and equal.
- **Rhombus** — 4 EQUAL sides (like a tilted square).
- **Trapezoid** — 4 sides, exactly ONE pair of parallel sides.
- **Pentagon** — 5 sides.
- **Hexagon** — 6 sides.
- **Octagon** — 8 sides (think: stop sign!).
- **Circle** — perfectly round, no straight sides or angles.

Perimeter — Distance Around a Shape

The **perimeter** is the total distance around the OUTSIDE of a shape. Think of it as how far an ant would walk if it traveled along every edge!

Formulas:

- **Rectangle:** $P = 2 \times \text{length} + 2 \times \text{width} = 2(l + w)$
- **Square:** $P = 4 \times \text{side} = 4s$
- **Triangle:** $P = \text{side} \blacksquare + \text{side} \blacksquare + \text{side} \blacksquare$
- **Any polygon:** Add up ALL the sides!

Units: Perimeter is measured in regular length units: inches (in), feet (ft), centimeters (cm), meters (m).

Area — Space Inside a Shape

The **area** is the amount of space INSIDE a flat shape. Think of it as how many square tiles would fit inside!

Formulas:

- **Rectangle:** $A = \text{length} \times \text{width} = l \times w$
- **Square:** $A = \text{side} \times \text{side} = s^2$

Units: Area is ALWAYS measured in SQUARE units: square inches (sq in or in²), square feet (sq ft or ft²), square centimeters (sq cm or cm²).

Important Difference:

- Perimeter = distance AROUND (like a fence)
- Area = space INSIDE (like carpet or paint)

Worked Examples: Geometry

Example 1: Find the perimeter of a rectangle: length=8 in, width=5 in

$$\text{Answer: } P = 2(l + w) = 2(8 + 5) = 2(13) = 26 \text{ inches}$$

Example 2: Find the area of a rectangle: length=8 in, width=5 in

$$\text{Answer: } A = l \times w = 8 \times 5 = 40 \text{ square inches}$$

Example 3: Find the perimeter of a square with side = 6 cm

$$\text{Answer: } P = 4 \times s = 4 \times 6 = 24 \text{ cm}$$

Example 4: Find the area of a square with side = 6 cm

$$\text{Answer: } A = s \times s = 6 \times 6 = 36 \text{ square cm}$$

Example 5: A garden is 12 ft long and 9 ft wide. Find its perimeter.

$$\text{Answer: } P = 2(12 + 9) = 2(21) = 42 \text{ feet. You need 42 ft of fencing!}$$

Example 6: Same garden: find its area.

$$\text{Answer: } A = 12 \times 9 = 108 \text{ square feet. That's how much soil you need!}$$

Example 7: A triangle has sides 5 cm, 7 cm, and 9 cm. Find the perimeter.

$$\text{Answer: } P = 5 + 7 + 9 = 21 \text{ cm}$$

Example 8: A square room has a side of 10 ft. How much carpet is needed?

Answer: $A = 10 \times 10 = 100$ square feet of carpet

Example 9: Find perimeter of rectangle: 15 m by 4 m

Answer: $P = 2(15 + 4) = 2(19) = 38$ m

Example 10: A picture frame is 11 in by 8 in. How much wood for the frame?

Answer: $P = 2(11 + 8) = 2(19) = 38$ inches of wood

Practice Problems: Geometry

Directions: Solve each geometry problem. Show your work.

1. How many sides does a hexagon have?
2. How many angles does a triangle have?
3. What shape has 4 equal sides and 4 right angles?
4. What shape looks like a stop sign? How many sides does it have?
5. Find the perimeter of a rectangle: length=10 in, width=7 in
6. Find the area of a rectangle: length=10 in, width=7 in
7. Find the perimeter of a square: side = 9 cm
8. Find the area of a square: side = 9 cm
9. A rectangle is 14 ft long and 6 ft wide. Find the perimeter.
10. Same rectangle: find the area.

Geometry Practice (continued)

11. A triangle has sides 8 cm, 6 cm, and 10 cm. Find the perimeter.
12. A square playground has sides of 25 m. Find the perimeter.
13. Same playground: find the area.
14. A rectangular garden is 20 ft by 15 ft. How much fence is needed?
15. Same garden: how much soil to cover the whole area?
16. A picture frame is 12 in by 9 in. Find the perimeter.
17. Find the area of a rectangle: 7 m by 11 m
18. A room is 13 ft by 10 ft. How much carpet is needed?
19. A rectangle has a perimeter of 30 cm. If the length is 10 cm, what is the width?
20. A square has an area of 64 sq in. What is the length of each side?

SECTION 6

LEAP 2025 Practice Test

LEAP 2025 Practice Test

Mixed Review — 30 Problems

Directions: Solve each problem. Show your work. Circle your final answer. You may use scratch paper.

Time Allowed: 45 minutes

1. $456 + 378 = ?$

2. $902 - 567 = ?$

3. $7 \times 8 = ?$

4. $63 \div 9 = ?$

5. What is $\frac{3}{4}$ of 20? (Hint: divide 20 by 4, then multiply by 3)

6. Find the perimeter of a rectangle that is 12 cm long and 8 cm wide.

7. Find the area of a square with sides of 7 feet.

8. $834 + 297 = ?$

9. $600 - 248 = ?$

$$10. 9 \times 12 = ?$$

LEAP Practice Test (continued)

11. $84 \div 7 = ?$

12. Simplify the fraction: $\frac{6}{9}$

13. Compare using $<$ or $>$: $\frac{2}{5}$ ____ $\frac{3}{10}$

14. A rectangle is 15 inches long and 9 inches wide. What is the area?

15. What is the perimeter of a triangle with sides 5 cm, 12 cm, and 13 cm?

16. $745 + 168 = ?$

17. $1,000 - 463 = ?$

18. $11 \times 6 = ?$

19. $96 \div 8 = ?$

20. Convert $\frac{11}{4}$ to a mixed number.

LEAP Practice Test (continued)

21. Convert $3\frac{2}{5}$ to an improper fraction.
22. A square field has sides of 30 meters. What is its perimeter?
23. Same field: what is its area?
24. $485 + 376 = ?$
25. $723 - 489 = ?$
26. $8 \times 7 = ?$
27. $52 \div 6 = ?$ (include the remainder)
28. Which fraction is larger: $\frac{2}{3}$ or $\frac{5}{8}$? Explain how you know.
29. A room is 14 ft by 11 ft. How many square feet of carpet are needed?
30. A bookshelf has 5 shelves. Each shelf holds 12 books. How many books total?

COMPLETE ANSWER KEY

Section 1: Addition & Subtraction

1. $804 + 264 = 1068$

2. $175 + 431 = 606$

3. $400 + 378 = 778$

4. $292 + 254 = 546$

5. $842 + 708 = 1550$

6. $239 + 754 = 993$

7. $582 + 182 = 764$

8. $180 + 245 = 425$

9. $373 + 388 = 761$

10. $667 + 766 = 1433$

11. $427 - 293 = 134$

12. $603 - 516 = 87$

13. $958 - 579 = 379$

14. $625 - 379 = 246$

15. $684 - 564 = 120$

16. $406 - 344 = 62$

17. $563 - 507 = 56$

18. $832 - 498 = 334$

19. $684 - 229 = 455$

20. $620 - 540 = 80$

Section 2: Multiplication

1. $3 \times 7 = 21$
2. $4 \times 8 = 32$
3. $6 \times 9 = 54$
4. $7 \times 7 = 49$
5. $8 \times 6 = 48$
6. $9 \times 4 = 36$
7. $5 \times 8 = 40$
8. $12 \times 3 = 36$
9. $11 \times 6 = 66$
10. $4 \times 9 = 36$
11. $6 \times 6 = 36$
12. $7 \times 9 = 63$
13. $8 \times 12 = 96$
14. $9 \times 9 = 81$
15. $5 \times 7 = 35$
16. $3 \times 11 = 33$
17. $6 \times 8 = 48$
18. $7 \times 4 = 28$
19. $12 \times 5 = 60$
20. $8 \times 7 = 56$

Section 3: Division

1. $48 \div 6 = 8$
2. $63 \div 7 = 9$
3. $81 \div 9 = 9$
4. $54 \div 6 = 9$
5. $96 \div 8 = 12$
6. $72 \div 9 = 8$
7. $108 \div 12 = 9$
8. $44 \div 4 = 11$
9. $60 \div 5 = 12$
10. $77 \div 7 = 11$
11. $29 \div 3 = 9 \text{ R}2$
12. $37 \div 4 = 9 \text{ R}1$
13. $50 \div 6 = 8 \text{ R}2$
14. $65 \div 8 = 8 \text{ R}1$
15. $43 \div 5 = 8 \text{ R}3$
16. $58 \div 7 = 8 \text{ R}2$
17. $83 \div 9 = 9 \text{ R}2$
18. $95 \div 11 = 8 \text{ R}7$
19. $47 \div 6 = 7 \text{ R}5$
20. $71 \div 8 = 8 \text{ R}7$

Section 4: Fractions

1. $\frac{5}{9}$
2. $\frac{7}{10}$
3. 3
4. 12
5. $\frac{3}{9}$
6. $\frac{6}{15}$
7. $\frac{15}{20}$
8. $\frac{3}{4}$
9. $\frac{1}{3}$
10. $\frac{2}{3}$
11. $\frac{3}{7} < \frac{5}{7}$
12. $\frac{1}{6} < \frac{1}{4}$
13. $\frac{2}{3} > \frac{3}{5}$ ($\frac{10}{15} > \frac{9}{15}$)
14. $2 \frac{1}{4}$
15. $3 \frac{2}{3}$
16. $\frac{7}{2}$
17. $\frac{12}{5}$
18. $\frac{15}{60} = \frac{1}{4}$
19. $\frac{4}{12} = \frac{1}{3}$
20. $\frac{5}{8}$

Section 5: Geometry

1. 6 sides
2. 3 angles
3. Square
4. Octagon, 8 sides
5. $P = 2(10+7) = 34$ in
6. $A = 10 \times 7 = 70$ sq in
7. $P = 4 \times 9 = 36$ cm
8. $A = 9 \times 9 = 81$ sq cm
9. $P = 2(14+6) = 40$ ft
10. $A = 14 \times 6 = 84$ sq ft
11. $P = 8+6+10 = 24$ cm
12. $P = 4 \times 25 = 100$ m
13. $A = 25 \times 25 = 625$ sq m
14. $P = 2(20+15) = 70$ ft fence
15. $A = 20 \times 15 = 300$ sq ft
16. $P = 2(12+9) = 42$ in
17. $A = 7 \times 11 = 77$ sq m
18. $A = 13 \times 10 = 130$ sq ft
19. $2(10+w)=30 \rightarrow 10+w=15 \rightarrow w=5$ cm
20. Side = 8 in (since $8 \times 8=64$)

Section 6: LEAP Practice Test

1. 834
2. 335
3. 56
4. 7
5. 15
6. $P = 2(12+8) = 40$ cm
7. $A = 7 \times 7 = 49$ sq ft
8. 1,131
9. 352
10. 108
11. 12
12. $\frac{2}{3}$
13. $\frac{2}{5} > \frac{3}{10}$ ($\frac{4}{10} > \frac{3}{10}$)
14. $A = 15 \times 9 = 135$ sq in
15. $P = 5+12+13 = 30$ cm
16. 913
17. 537
18. 66
19. 12
20. $2 \frac{3}{4}$
21. $\frac{17}{5}$
22. $P = 4 \times 30 = 120$ m
23. $A = 30 \times 30 = 900$ sq m
24. 861
25. 234
26. 56
27. 8 R4
28. $\frac{2}{3} > \frac{5}{8}$ ($\frac{16}{24} > \frac{15}{24}$)
29. $A = 14 \times 11 = 154$ sq ft
30. $5 \times 12 = 60$ books